

Tyler Hobbs

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EDUCATION

University of Virginia

Master's in Data Science

Charlottesville, VA

Jan. 2025 – Dec. 2026

University of Colorado

Bachelor of Arts in Molecular, Cellular, and Developmental Biology

Boulder, CO

Aug. 2018 – Dec. 2021

TECHNICAL SKILLS

Languages: Python, SQL, R, HTML/CSS

Statistics & Modeling: Hypothesis Testing, Regression, Cross-Validation, Class Imbalance Methods, Bayesian Inference, Time Series Analysis

ML/DL Frameworks: PyTorch, TensorFlow, scikit-learn, tidymodels, FinRL, Hugging Face

ML Techniques: Transformers, CNNs, Semantic Segmentation, Reinforcement Learning (PPO), LoRA/PEFT, XGBoost, Random Forests, LDA/QDA, Elastic Net, FinBERT

Data & Visualization: pandas, NumPy, Plotly/Dash, D3.js, Matplotlib, Power BI, Tableau, Excel

Tools & Infrastructure: Git, Flask, Docker, PostgreSQL, MongoDB, HPC (SLURM), REST APIs

TECHNICAL PROJECTS

Glaucoma Segmentation — Multi-Model OD/OC | *PyTorch, SAM/PEFT, HPC*

June 2026 – Aug. 2026

- Built a reproducible, manifest-driven pipeline comparing U-Net, U-Net++, DeepLabV3+ for joint optic-disc/cup segmentation on pooled fundus datasets (ORIGA, G1020, REFUGE, and PAPILA).
- Engineered a shared training/eval harness where identical Dice+CE loss and per-image Dice/IoU/CDR metrics, 3 seeds per model so architecture is the only variable across runs.
- Held out the PAPILA clinical dataset for zero-shot domain-transfer evaluation, using vertical cup-to-disc ratio (CDR) error as the clinical screening endpoint.
- Awarded **Most Innovative Analytical Solution** for model novelty and performance against client requirements; shipped a Hugging Face Spaces demo, manuscript, and public repository to the UVA Ophthalmology client team.

LoadBrief — LLM Fine-Tuning for Athlete Load Management | *Python, Llama 3, LoRA, GRPO*

June 2026 – Present

- Fine-tuned Llama 3 8B Instruct (LoRA SFT) to convert athlete monitoring narratives into audience-conditioned load-management briefs across three registers (athlete, coach, sports scientist), training locally on a 48GB M4 across four SFT iterations at ~32 h per cycle.
- Rebuilt the 50K-example synthetic corpus behind a deterministic consistency checker, driving the defect rate from 8.8% to 0.1% and isolating six scenario classes whose targets were unreachable by construction; published as a Hugging Face dataset and model card.
- Lifted risk-classification accuracy from 0.70 to 0.79 via greedy decoding, reaching within-one-class accuracy of 0.96–0.98 with zero catastrophic (low↔critical) misclassifications.
- Found all 8 GRPO configurations null against the SFT baseline due to reward saturation by construction; calibrated the LLM judge against hand-written reference briefs to confirm the ceiling was a corpus artifact rather than judge miscalibration. Manuscript in preparation.

FinRL Deep Reinforcement Learning | *Python, PyTorch, PPO, FinBERT, Alpaca API*

Feb. 2026 – Present

- Designed a 24-model PPO ablation study within the FinRL framework, benchmarking CNN and cross-stock Transformer architectures across 30/50-stock universes and three capital levels.
- Conducted multi-seed robustness analysis (3 seeds × 4 architectures), finding seed initialization the dominant variable and no configuration reliably beating buy-and-hold; a peak Sharpe of 2.531 on a truncated window did not persist under full out-of-sample testing.
- Integrated real-time Polygon.io news sentiment via FinBERT into the agent's state; deployed to live paper trading on the Alpaca API with automated risk controls and a Flask monitoring dashboard.

EXPERIENCE

Clinical Laboratory Technician II

Feb. 2026 – Present

Illumina Inc.

Boulder, CO

- Replaced legacy Power BI dashboard with a Python/HTML web app, surfacing per-plate-run scanner metrics not previously tracked and supporting Performance Qualification workflows across all laboratory scanners.
- Built an interactive dashboard ingesting Salesforce opportunity data and weekly lab priorities, presenting daily priorities, opportunity-level metrics, and per-study timelines across three tabs — displayed on a lab TV for the Sample Management team.
- Developed a companion ETL pipeline that processes raw Salesforce exports (footer stripping, merged-ID forward-filling, per-status-change timestamp reconstruction) to produce structured event-log data for downstream analysis.

Laboratory Associate II

Mar. 2022 – Feb. 2026

Standard Biotools Inc. (formerly SomaLogic Inc.)

Boulder, CO

- Led daily equipment maintenance and trained new team members on the Tecan Fluent 780, reducing non-conformance events by nearly 10%.
- Served as lead maintenance specialist for Tecan Fluent liquid handlers and Fluent Control software, maintaining a 2-year streak of zero assay failures attributable to equipment issues.